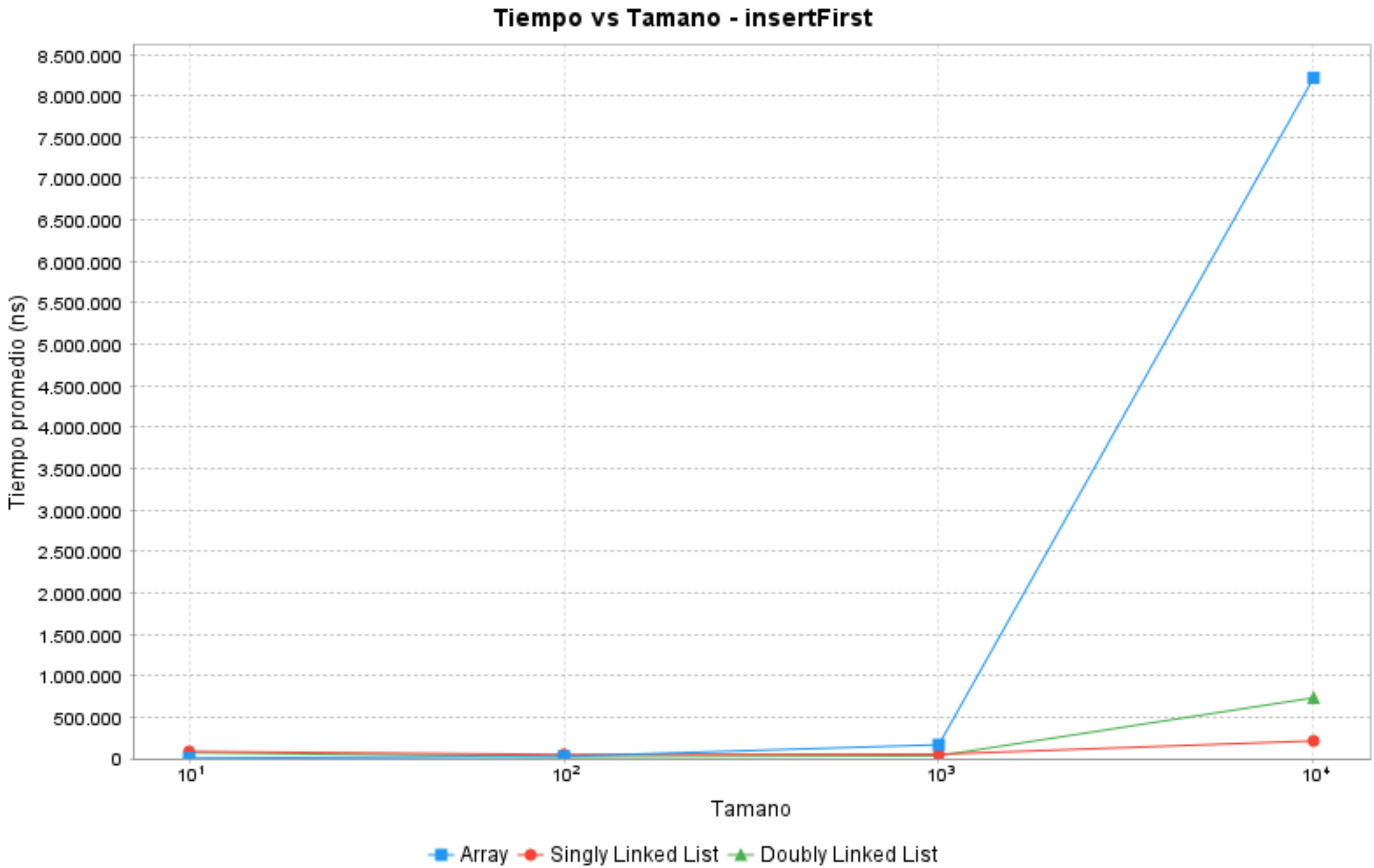
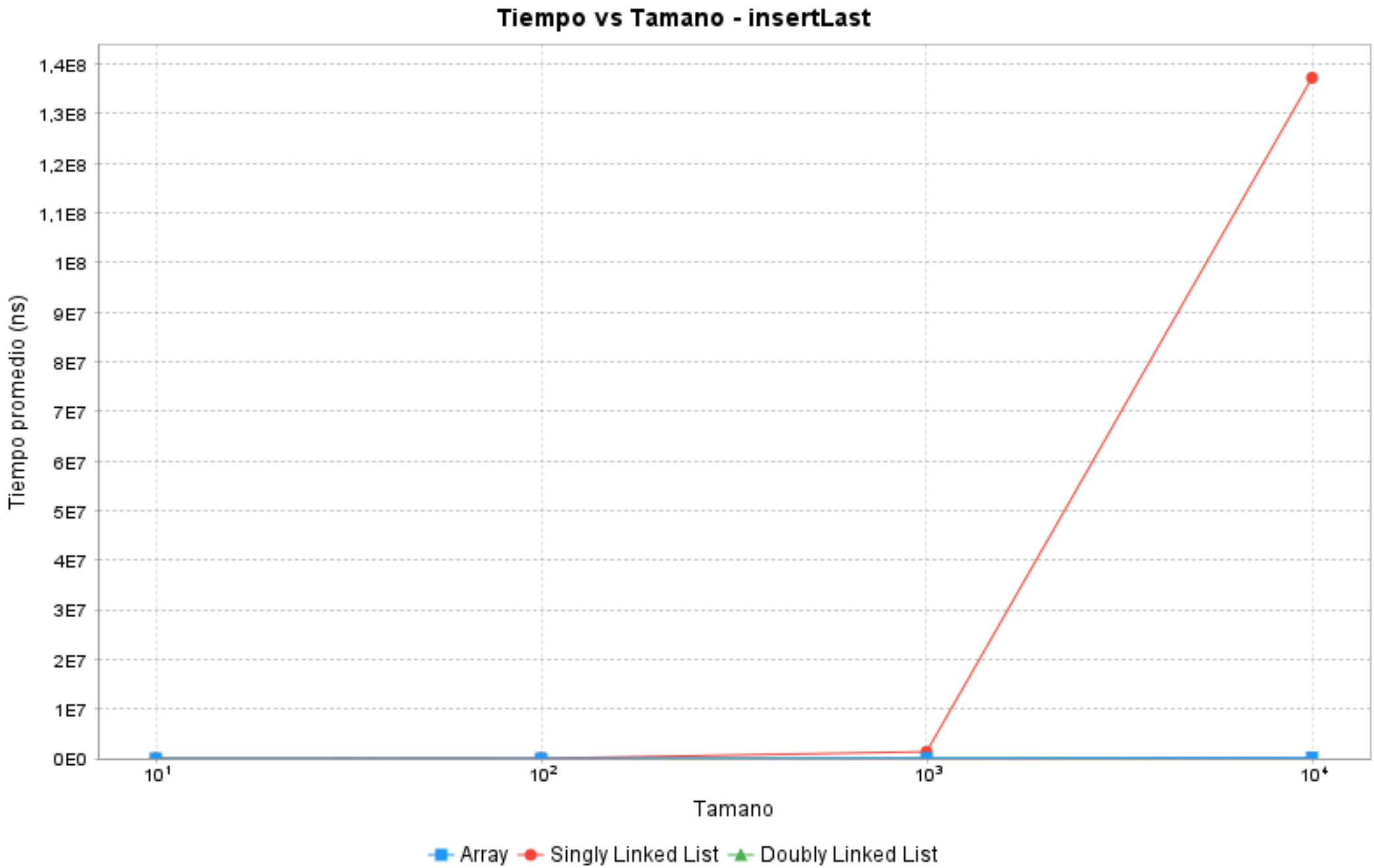
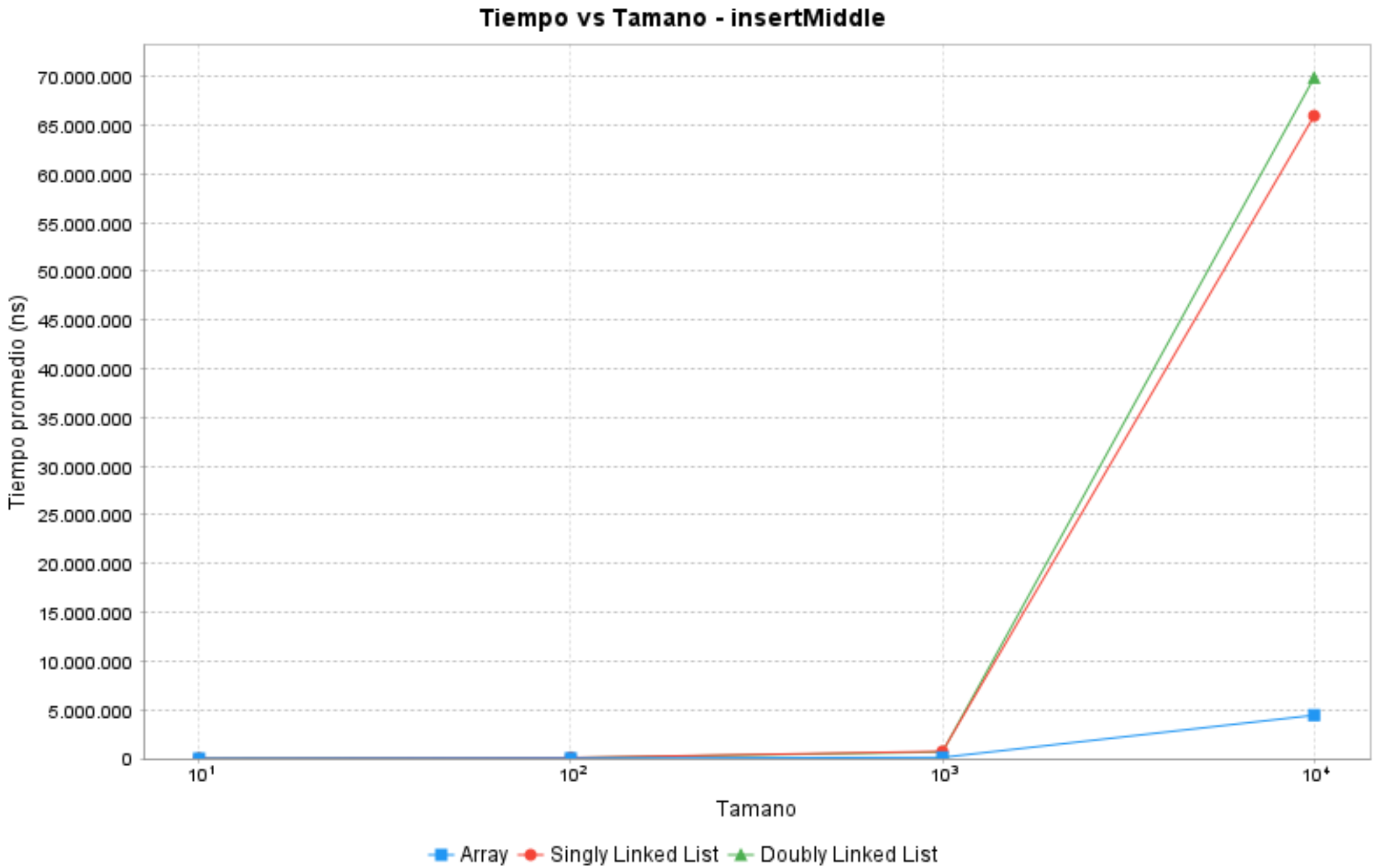
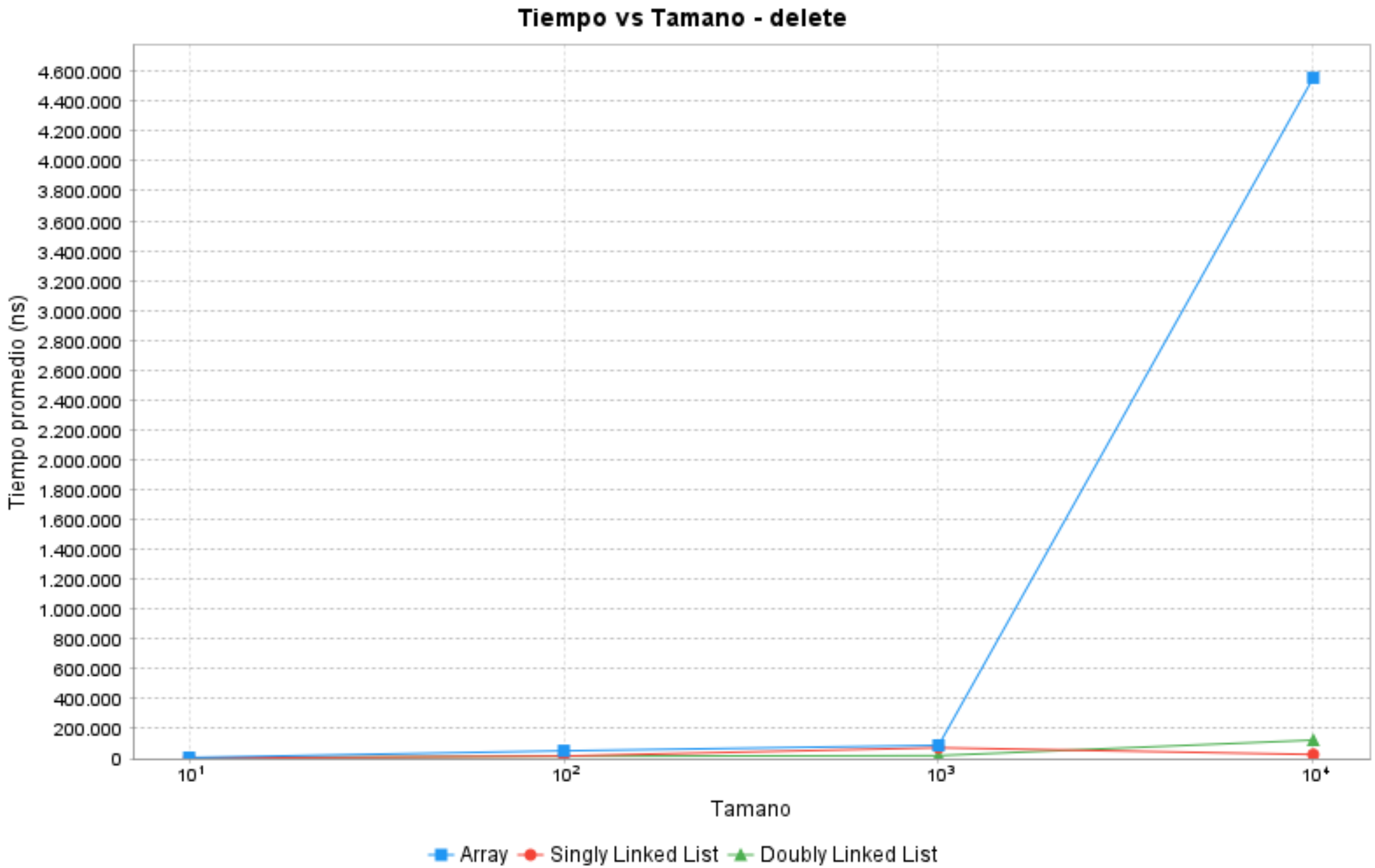
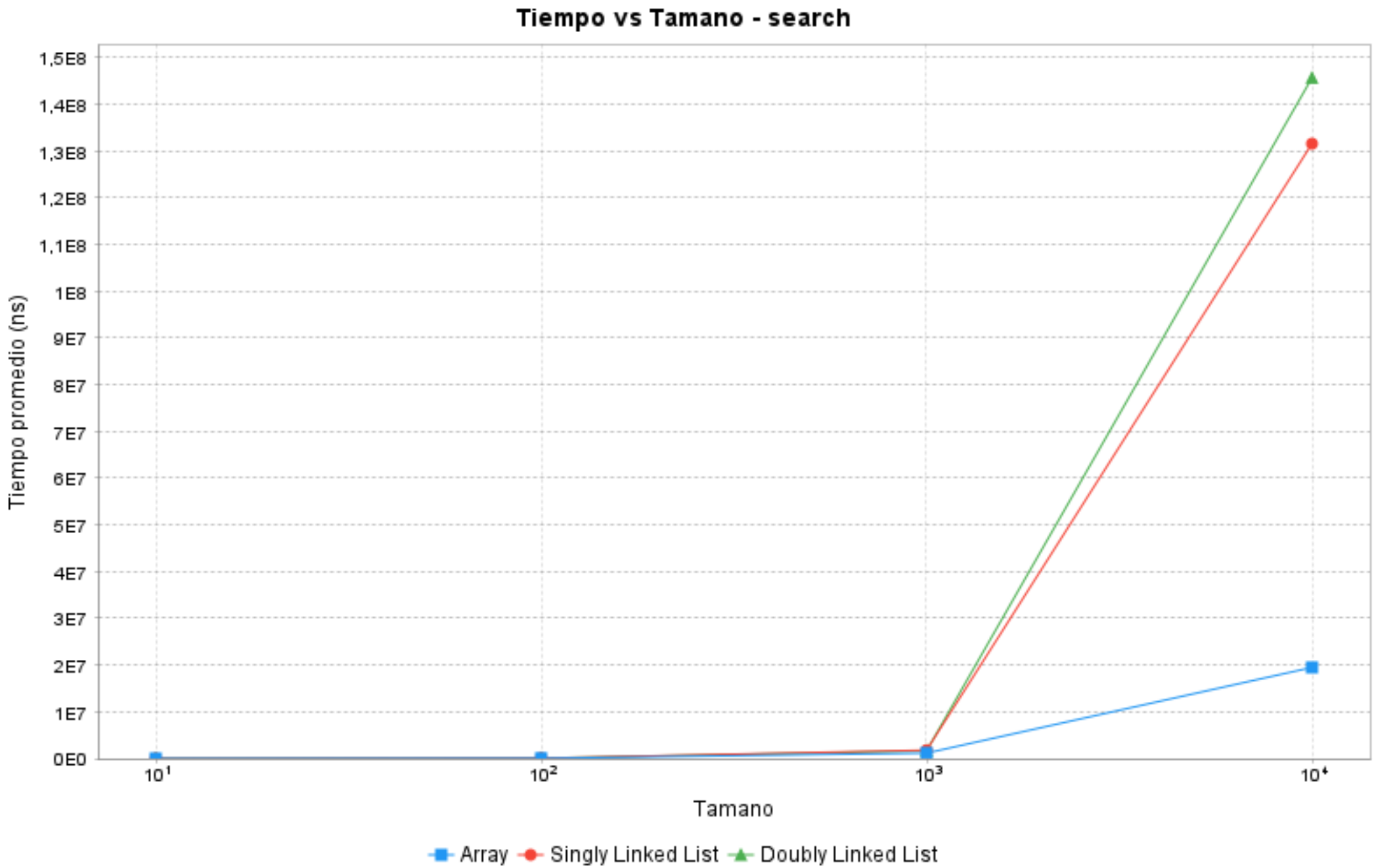
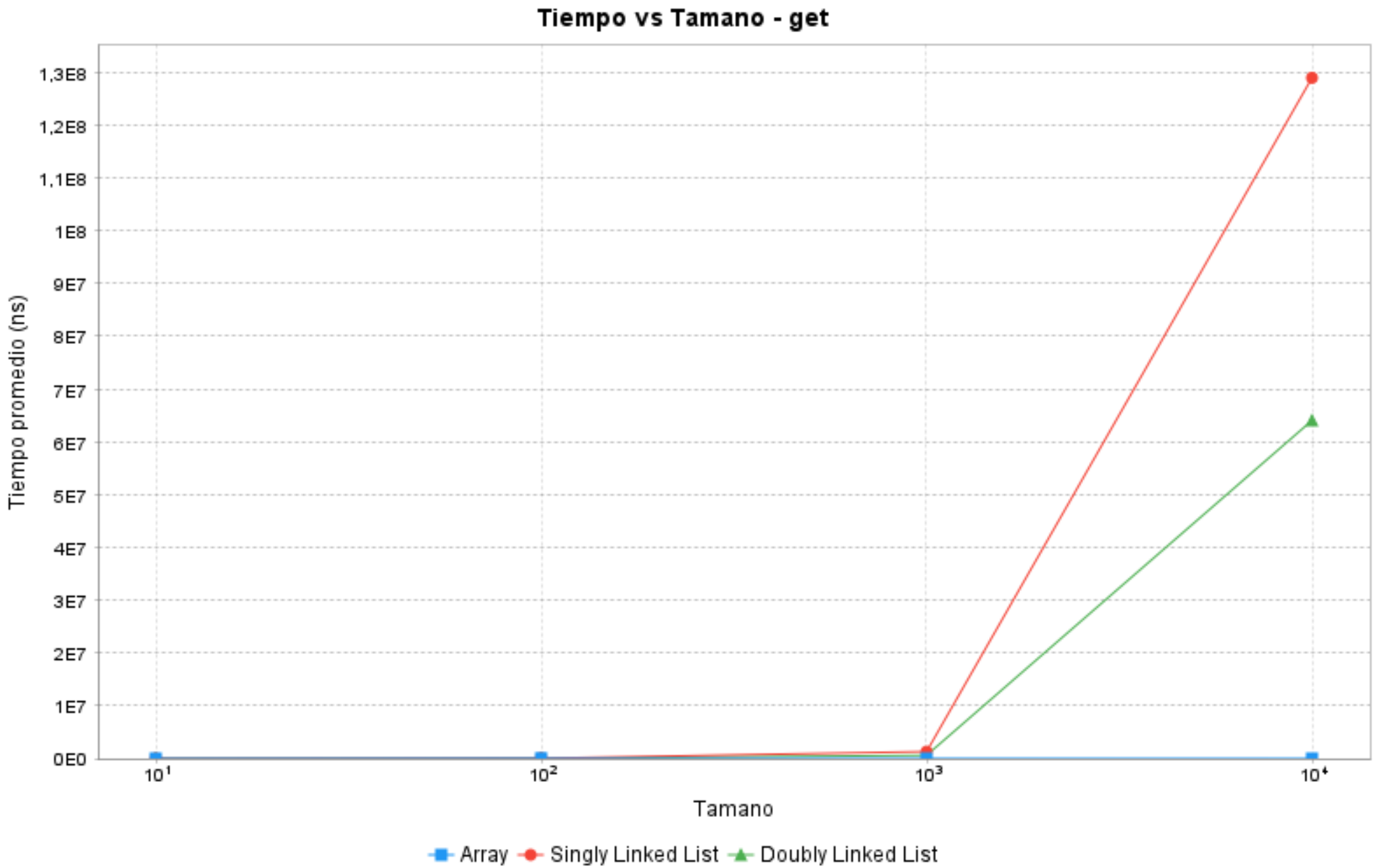

Tiempo vs Tamano - insertFirst
Tiempo promedio (ns)
8.500.000
8.000.000
7.500.000
7.000.000
6.500.000
6.000.000
5.500.000
5.000.000
4.500.000
4.000.000
3.500.000
3.000.000
2.500.000
2.000.000
1.500.000
1.000.000
500.000
0
10^1
10^2
10^3
10^4
Tamano
Array
Singly Linked List
Doubly Linked List

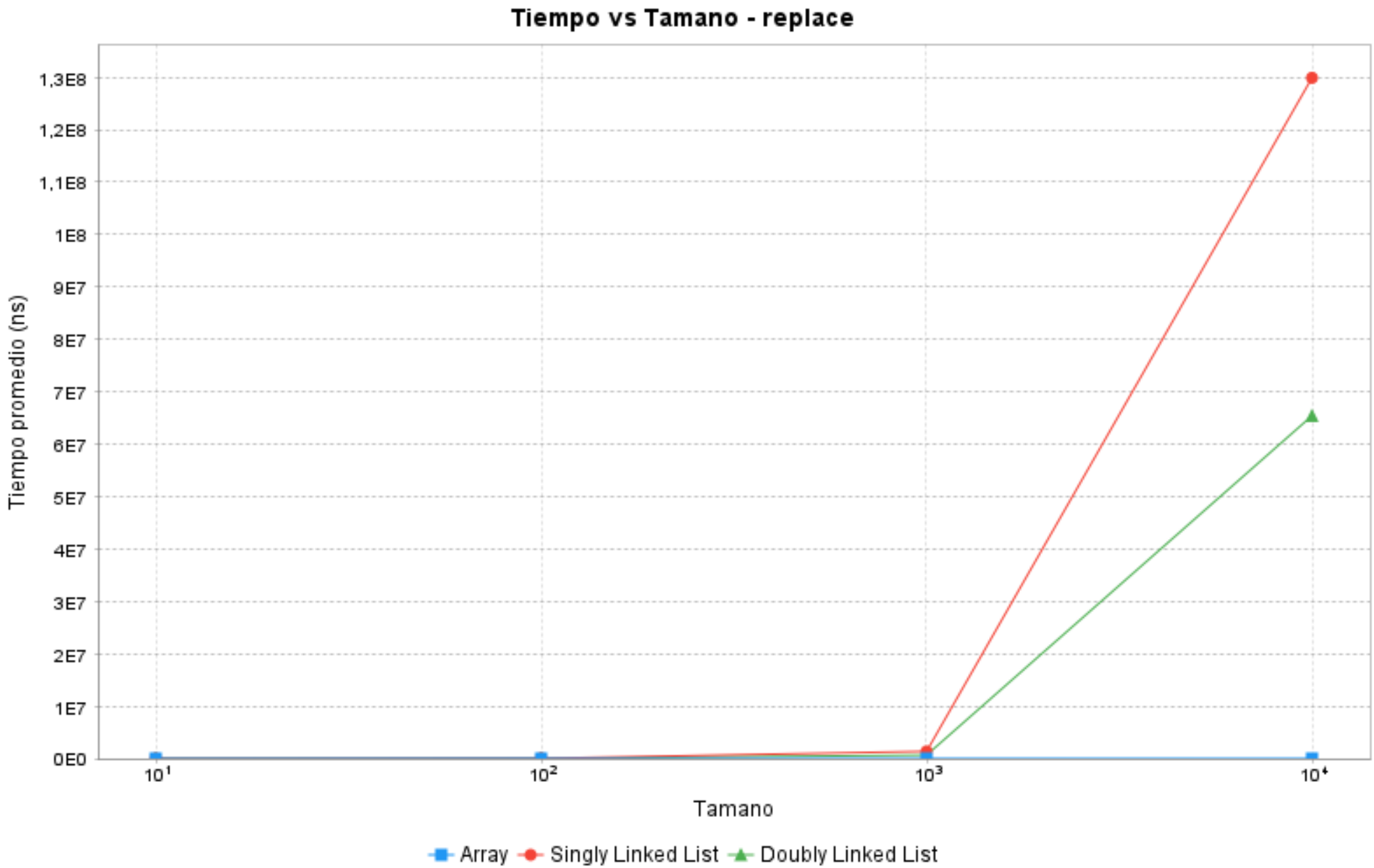

Tiempo vs Tamano - insertLast
Tiempo promedio (ns)
1,4E8
1,3E8
1,2E8
1,1E8
1E8
9E7
8E7
7E7
6E7
5E7
4E7
3E7
2E7
1E7
0E0
10^1
10^2
10^3
10^4
Tamano
Array
Singly Linked List
Doubly Linked List

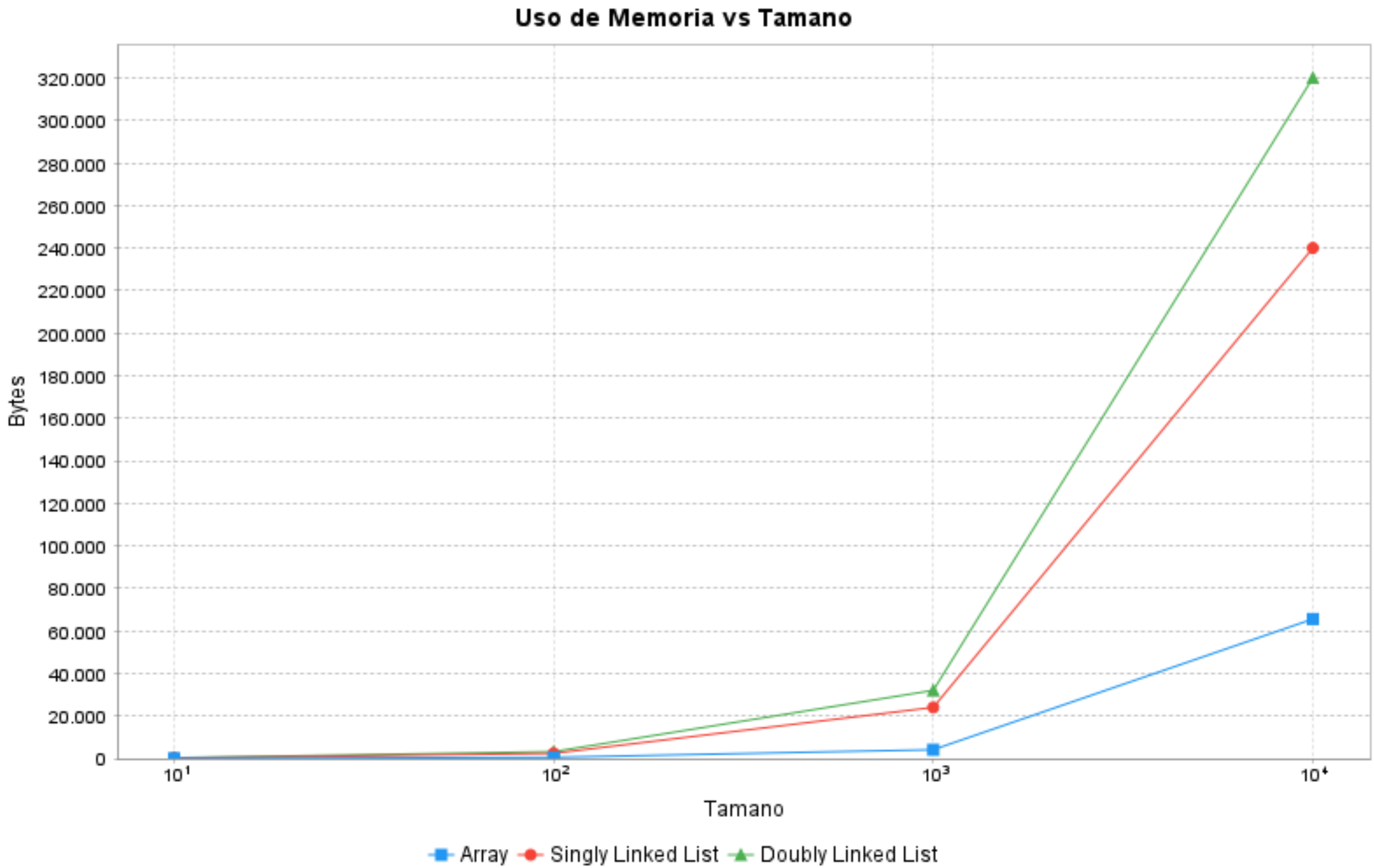

Tiempo vs Tamano - insertMiddle
Tiempo promedio (ns)
70.000.000
65.000.000
60.000.000
55.000.000
50.000.000
45.000.000
40.000.000
35.000.000
30.000.000
25.000.000
20.000.000
15.000.000
10.000.000
5.000.000
0
10^1
10^2
10^3
10^4
Tamano
Array
Singly Linked List
Doubly Linked List

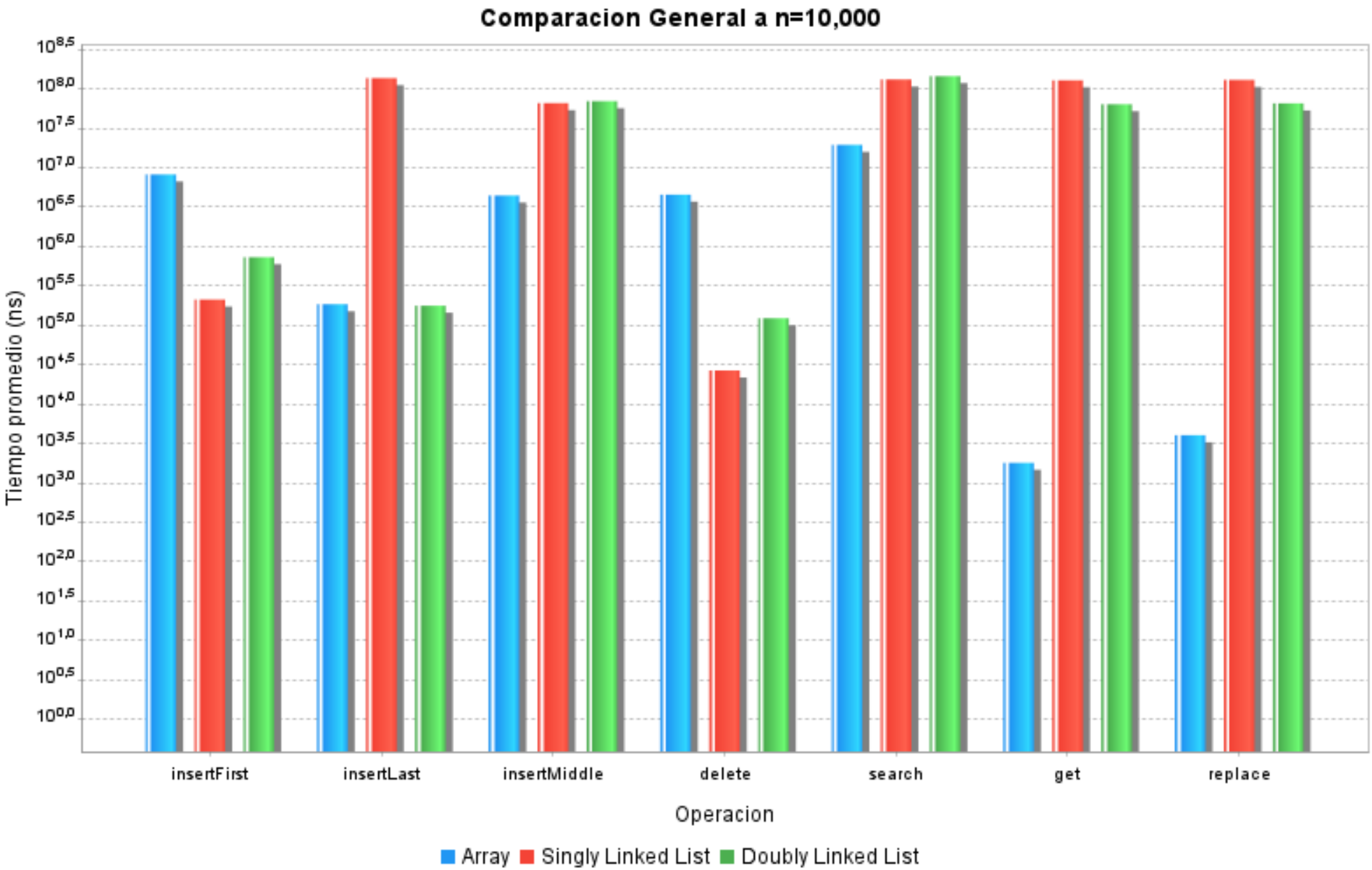

Tiempo vs Tamano - delete
Tiempo promedio (ns)
4.600.000
4.400.000
4.200.000
4.000.000
3.800.000
3.600.000
3.400.000
3.200.000
3.000.000
2.800.000
2.600.000
2.400.000
2.200.000
2.000.000
1.800.000
1.600.000
1.400.000
1.200.000
1.000.000
800.000
600.000
400.000
200.000
0
10^1
10^2
10^3
10^4
Tamano
Array
Singly Linked List
Doubly Linked List


Tiempo vs Tamano - search
Tiempo promedio (ns)
1,5E8
1,4E8
1,3E8
1,2E8
1,1E8
1E8
9E7
8E7
7E7
6E7
5E7
4E7
3E7
2E7
1E7
0E0
10^1
10^2
10^3
10^4
Tamano
Array
Singly Linked List
Doubly Linked List


Tiempo vs Tamano - get
Tiempo promedio (ns)
1,3E8
1,2E8
1,1E8
1E8
9E7
8E7
7E7
6E7
5E7
4E7
3E7
2E7
1E7
0E0
10^1
10^2
10^3
10^4
Tamano
Array
Singly Linked List
Doubly Linked List


Tiempo vs Tamano - replace
Tiempo promedio (ns)
1,3E8
1,2E8
1,1E8
1E8
9E7
8E7
7E7
6E7
5E7
4E7
3E7
2E7
1E7
0E0
10^1
10^2
10^3
10^4
Tamano
Array
Singly Linked List
Doubly Linked List


Uso de Memoria vs Tamano
Bytes
320.000
300.000
280.000
260.000
240.000
220.000
200.000
180.000
160.000
140.000
120.000
100.000
80.000
60.000
40.000
20.000
0
10^1
10^2
10^3
10^4
Tamano
Array
Singly Linked List
Doubly Linked List


Comparacion General a n=10,000
Tiempo promedio (ns)
$10^{8,5}$
$10^{8,0}$
$10^{7,5}$
$10^{7,0}$
$10^{6,5}$
$10^{6,0}$
$10^{5,5}$
$10^{5,0}$
$10^{4,5}$
$10^{4,0}$
$10^{3,5}$
$10^{3,0}$
$10^{2,5}$
$10^{2,0}$
$10^{1,5}$
$10^{1,0}$
$10^{0,5}$
$10^{0,0}$
insertFirst
insertLast
insertMiddle
delete
search
get
replace
Operacion
Array
Singly Linked List
Doubly Linked List